1. Standard Test Signals

The actual input to a system is often unknown, so its performance is compared using standard test signals whose Laplace transforms are simple.

Signal	r(t)	R(s)	Used to test
Step	$A\,u(t)$	A/s	Sudden change (position); transient specs
Ramp	$A\,t$	A/s^2	Constant-velocity inputs (tracking)
Parabolic	$A\,t^2/2$	A/s^3	Constant acceleration
Impulse	$A\,\delta(t)$	A	Sudden shock; gives the T.F. (impulse response)
Sinusoidal	$A \sin \omega t$	$A\omega/(s^2+\omega^2)$	Frequency response

Time response $c(t)$ = **transient** response (dies out as $t \to \infty$) + **steady-state** response (remains as $t \to \infty$).

2. Time Response of First Order Systems

$C(s)/R(s) = 1 / (T s + 1)$ (T = time constant, pole at $s = -1/T$)

Input	C(s)	c(t)	Remarks
Unit step	$1/[s(Ts+1)]$	$1 - e^{-t/T}$	$c(T) = 63.2\ \%$ of final value; no overshoot
Unit ramp	$1/[s^2(Ts+1)]$	$t - T + T\,e^{-t/T}$	$e_{ss} = T$ (lags the input)
Unit impulse	$1/(Ts+1)$	$(1/T)\,e^{-t/T}$	Decays to zero; derivative of step response

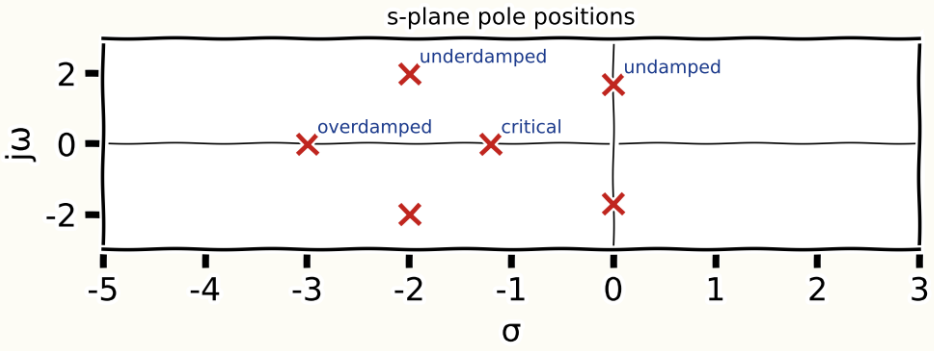
- **Time constant T:** time taken to reach 63.2 % of the final value for a step input.
- Settling time: $t_s = 4T$ (2 % band), $3T$ (5 % band); rise time t_r (10 %–90 %) = $2.2T$; delay time $t_d = 0.69T$.
- Smaller $T \rightarrow$ pole farther left $\rightarrow$ faster response. Response to derivative of input = derivative of response (LTI property).

3. Characteristic Equation of Feedback Control Systems

For the closed loop system $C/R = G/(1 + GH)$ the denominator set to zero is the **characteristic equation**:

$$\mathbf{1 + G(s)H(s) = 0 \Rightarrow roots = closed\ loop\ poles}$$

- The poles decide stability and nature of the transient response; zeros affect only the magnitude/shape.
- Poles in LHP $\rightarrow$ stable decaying terms; on imaginary axis $\rightarrow$ sustained oscillation; in RHP $\rightarrow$ growing.



4. Transient Response of Second Order Systems

$$C(s)/R(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$$

ω_n = natural frequency, ζ = damping ratio, $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$ = damped frequency

Poles: $s = -\zeta\omega_n \pm j\omega_n\sqrt{(1-\zeta^2)}$

For unity feedback with $G(s) = \omega_n^2/[s(s + 2\zeta\omega_n)]$ the closed loop gives the above T.F.

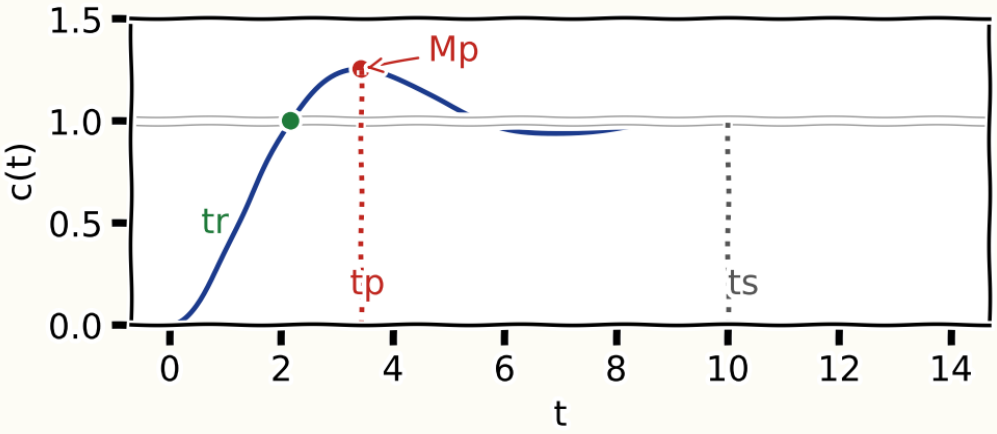
ζ	Type	Poles	Step response
0	Undamped	$\pm j\omega_n$ (imag. axis)	Sustained oscillations (constant amplitude)
$0 < \zeta < 1$	Underdamped	complex conjugate, LHP	Damped oscillations, overshoot
1	Critically damped	real, equal ($-\omega_n$)	Fastest non-oscillatory rise
> 1	Overdamped	real, unequal	Sluggish, no overshoot

Step response – underdamped ($0 < \zeta < 1$)

$$c(t) = 1 - [\, e^{-\zeta\omega_n t} / \sqrt{(1-\zeta^2)}\,] \sin(\omega_d t + \varphi)\,,\ \varphi = \cos^{-1}\zeta = \tan^{-1}[\sqrt{(1-\zeta^2)}/\zeta]$$

Critically damped: $c(t) = 1 - (1 + \omega_n t)\, e^{-\omega_n t}$. Overdamped: $c(t) = 1 + [1/(2\sqrt{(\zeta^2-1)}(\zeta+\sqrt{(\zeta^2-1)}))]\, e^{-(\zeta-\sqrt{(\zeta^2-1)})\omega_n t} - [1/(2\sqrt{(\zeta^2-1)}(\zeta-\sqrt{(\zeta^2-1)}))]\, e^{-(\zeta+\sqrt{(\zeta^2-1)})\omega_n t}$.

5. Time Domain Specifications



Specification	Definition	Formula (underdamped)
Delay time t_d	Time to reach 50 % of final value	$\approx (1 + 0.7\zeta)/\omega_n$
Rise time t_r	0→100 % (underdamped)	$t_r = (\pi - \varphi)/\omega_d$
Peak time t_p	Time of first peak	$t_p = \pi/\omega_d$
Peak overshoot M_p	$(c_{max} - c_{ss})/c_{ss} \times 100$	$M_p = e^{-\pi\zeta/\sqrt{(1-\zeta^2)}} \times 100$ %
Settling time t_s	Time to stay within 2 % (5 %) band	$t_s = 4/(\zeta\omega_n)$ $(3/(\zeta\omega_n))$

$$\zeta \text{ from } M_p:\ \zeta = -\ln(M_p/100) / \sqrt{(\pi^2 + \ln^2(M_p/100))}$$

Solved example

Unity feedback, $G(s) = 25/[s(s+5)]$. Closed loop: $\omega_n^2 = 25 \rightarrow \omega_n = 5$, $2\zeta\omega_n = 5 \rightarrow \zeta = 0.5$, $\omega_d = 5\sqrt{0.75} = 4.33$ rad/s, $\varphi = \cos^{-1}0.5 = 1.047$ rad.

$$t_r = (\pi - 1.047)/4.33 = 0.484\ s\ ;\ t_p = \pi/4.33 = 0.725\ s$$
$$M_p = e^{-\pi(0.5)/0.866} = 16.3\ \%\ ;\ t_s\ (2\ \%) = 4/(0.5\times5) = 1.6\ s$$

★ Higher $\zeta \rightarrow$ lower overshoot; higher ω_n (at same ζ) $\rightarrow$ faster response (smaller t_r, t_p, t_s). Desirable $\zeta = 0.4 - 0.707$.

6. Steady State Response and Steady State Errors

Steady-state error is the difference between input and output as $t \rightarrow \infty$. For unity feedback, $E(s) = R(s)/[1 + G(s)]$, by final value theorem:

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to0} s\,R(s) / [\,1 + G(s)H(s)\,]$$

System type and error constants

Write $G(s)H(s) = K(1+T_a s)\ldots / [\,s^N(1+T_1 s)\ldots\,]$; N = number of poles at origin = **type** of the system.

$$\text{Position error constant } K_p = \lim_{s\to0} G(s)H(s)$$
$$\text{Velocity error constant } K_v = \lim_{s\to0} s\,G(s)H(s)$$
$$\text{Acceleration error constant } K_a = \lim_{s\to0} s^2\,G(s)H(s)$$

Type	Step $e_{ss}=1/(1+K_p)$	Ramp $e_{ss}=1/K_v$	Parabolic $e_{ss}=1/K_a$
0	$1/(1+K)$	∞	∞
1	0	$1/K$	∞
2	0	0	$1/K$
3	0	0	0

Solved: $G(s)H(s) = 10(s+2)/[s^2(s+1)(s+5)]$ → *Type 2,* $K_p = \infty$, $K_v = \infty$, $K_a = 10 \cdot 2/(1 \cdot 5) = 4$ → e_{ss} *(step) = 0, (ramp) = 0, (parabolic with unit magnitude) =* $1/4 = 0.25$.

Effects of P-D and P-I controllers on the time response

- ***Proportional-Derivative (PD):*** $G_c = K_p(1 + T_d s)$ *adds a zero →* *increases damping* ($\zeta_{eff} = \zeta + \omega_n T_d/2$ *for the standard* 2nd*-order loop), reduces* M_p, t_r, t_s*; steady-state error for ramp unchanged (type unchanged); amplifies noise.*
- ***Proportional-Integral (PI):*** $G_c = K_p(1 + 1/(T_i s))$ *adds a pole at origin →* ***type increases by 1****, steady-state error reduces (step error becomes zero for type 0); but system becomes less stable and more oscillatory (*M_p *increases), slower response.*

7. Stability Analysis – Concept of Stability

A linear time-invariant system is ***stable*** *if for a bounded input the output is bounded (BIBO) and for zero input the output returns to zero from any initial condition.*

Condition	Location of closed-loop poles	Impulse response
Absolutely stable	All in left half s-plane	→ 0
Marginally (critically) stable	Simple poles on jω axis, none in RHP	sustained oscillation / constant
Unstable	Any pole in RHP or repeated poles on jω axis	grows without bound
Conditionally stable	Stable only for a limited range of gain K	–

- **Necessary condition:** all coefficients of the characteristic polynomial must be real, positive and non-zero (none missing). Not sufficient for order ≥ 3.
- **Relative stability:** how far the poles are from the jω axis (measured by shifting $s \to s - \sigma$, or gain/phase margins).

8. Routh–Hurwitz Stability Criterion

For $q(s) = a_0 s^n + a_1 s^{n-1} + \ldots + a_n = 0$ form the Routh array. **The system is stable iff all elements of the first column have the same sign**; the number of sign changes = number of roots in the RHP.

s^n	a_0	a_2	a_4
s^{n-1}	a_1	a_3	a_5
s^{n-2}	$b_1 = (a_1a_2 - a_0a_3)/a_1$	$b_2 = (a_1a_4 - a_0a_5)/a_1$	…
s^{n-3}	$c_1 = (b_1a_3 - a_1b_2)/b_1$	…	…
…	…	…	…
s^0	a_n		

Example 1: $s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$

s^4	1	3	5
s^3	2	4	0
s^2	$(6-4)/2 = 1$	$(10-0)/2 = 5$	0
s^1	$(4-10)/1 = -6$	0	0
s^0	5		

Signs: + + + − + → 2 sign changes → ***2 roots in RHP, unstable.***

Example 2: range of K for $s^3 + 6s^2 + 8s + K = 0$

Array: s^3: 1, 8 | s^2: 6, K | s^1: (48 − K)/6 | s^0: K. For stability K > 0 and 48 − K > 0:

0 < K < 48 (marginal at K = 48: $A(s) = 6s^2 + 48 = 0 \Rightarrow s = \pm j2.83$, so $\omega_{osc} = 2.83$ rad/s)

Special cases

- ***Case 1 – a zero appears in the first column*** *(other row elements non-zero): replace 0 by a small +ε, complete the array, then let ε → 0 and count sign changes. (Or substitute s = 1/z.)*

- **Case 2 – an entire row becomes zero**: *form the auxiliary equation A(s) from the row above, differentiate dA/ds, put its coefficients in the zero row, continue. Roots of A(s) are symmetrical about the origin ($\pm\sigma$ or $\pm j\omega$ pairs); if no sign change below, they are on $j\omega$ axis → marginally stable.*

Conditional stability

Some systems are stable only for a certain range of gain (e.g., $0 < K < 48$ above) or become stable only when K exceeds a minimum (e.g., higher-order systems stable for $K_1 < K < K_2$). Such a system is ***conditionally stable****; reducing gain below K_1 (for example due to saturation) can make it unstable.*

> ★ ***Limitations of Routh criterion: (1) applicable only to polynomials with real, constant coefficients (no transport delay e^{-sT} unless approximated); (2) gives no exact root locations; (3) does not indicate relative stability or how to stabilise the system; (4) not directly for non-polynomial/ irrational equations.***

9. Root Locus Concept

The ***root locus*** *is the path traced by closed-loop poles in the s-plane as a parameter (usually gain K, $0 \rightarrow \infty$) varies.*

Characteristic equation $1 + K\,G(s)H(s) = 0$ gives conditions:

Angle condition: $\angle G(s)H(s) = \pm(2q+1)180^\circ$, $q = 0, 1, 2, \ldots$

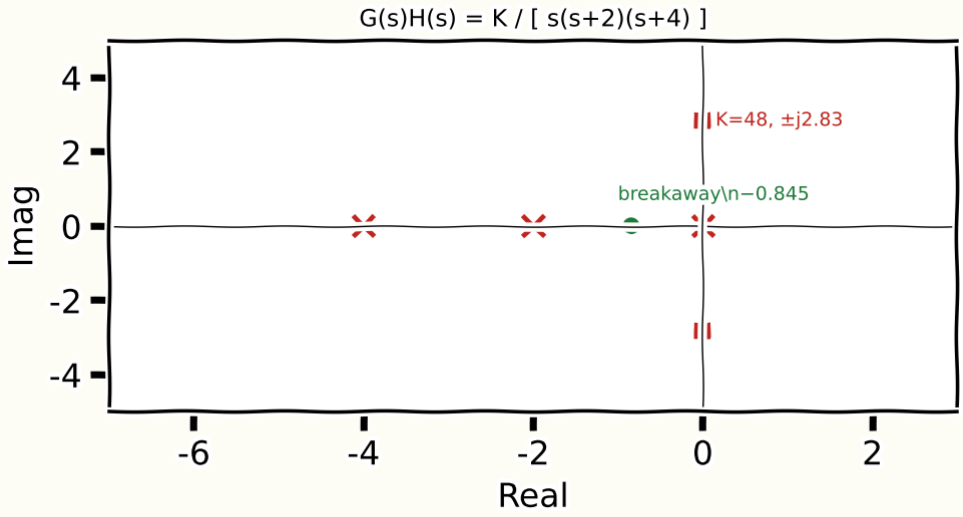
Magnitude condition: $|K\,G(s)H(s)| = 1 \Rightarrow K = \prod|s - \text{poles}| \,/\, \prod|s - \text{zeros}|$

Construction rules (P = no. of open loop poles, Z = no. of zeros)

#	Rule
1	Locus starts ($K=0$) at open loop poles and ends ($K=\infty$) at zeros; $P-Z$ branches go to infinity (zeros at infinity). No. of branches = $\max(P, Z)$.
2	Locus is symmetric about the real axis.
3	A point on the real axis lies on the locus if the number of poles + zeros to its right is **odd**.
4	Asymptotes: angles $\theta = (2q+1)180^\circ/(P-Z)$; centroid $\sigma_A = (\Sigma \text{ poles} - \Sigma \text{ zeros})/(P - Z)$.
5	Breakaway / break-in points: solve $dK/ds = 0$ with $K = -1/[G(s)H(s)]$; accept real roots lying on the locus.
6	Angle of departure from complex pole: $180^\circ - \Sigma$(angles of other poles) + Σ(angles of zeros); angle of arrival similarly.
7	Imaginary axis crossing: from Routh array (K_{max} with row of zeros) → auxiliary equation gives ω.
8	Gain K at any point s_1 on the locus from the magnitude condition.

Example: G(s)H(s) = K / [s(s+2)(s+4)]

- $P = 3, Z = 0 \rightarrow 3$ branches, all go to ∞. Real axis: $[0, -2]$ and $(-4, -\infty)$.
- Asymptotes: $\theta = 60°, 180°, 300°$; centroid $\sigma_A = (0 - 2 - 4)/3 = -2$.
- Breakaway: $K = -(s^3 + 6s^2 + 8s)$; $dK/ds = 0 \Rightarrow 3s^2 + 12s + 8 = 0 \Rightarrow s = -0.845$ (valid), -3.155 (not on locus).
- Crossing: Routh gives $K = 48$, $6s^2 + 48 = 0 \Rightarrow s = \pm j2.83$. System stable for $0 < K < 48$.



Effect of adding poles and zeros to G(s)H(s) on the root locus

Addition	Effect on root locus	Effect on system
Pole (LHP)	Locus bends/shifts to the **right**; asymptote angles change; centroid moves right	Reduces relative stability, increases oscillation; K_{max} decreases
Pole at origin	Increases type; locus moves right	Better steady-state accuracy, poorer stability
Zero (LHP)	Locus bends to the **left** (toward the zero); centroid moves left	Increases stability and damping; faster response
Zero in RHP	Moves locus to right	Reduces stability (non-minimum phase)

Important Questions

- Derive expressions for t_r, t_p, M_p, t_s of an underdamped second order system. (10M)
- Explain the step response of a second-order system for various damping ratios. (5M)
- Define K_p, K_v, K_a and find steady-state errors for a given system. (10M)
- State Routh's criterion; special cases with examples; find K for stability. (10M)
- Sketch the root locus for the given G(s)H(s); find breakaway, $j\omega$-crossing and K_{max}. (10M)
- Effect of adding poles and zeros on root locus; effects of PD and PI controllers. (5M)